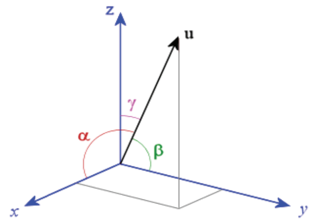
$\| \vec{v} \| = \sqrt{((-1)^2 + 0^2 + (3)^2)} = \sqrt{10} \neq 1$

In the same way, for a given direction (α, β, γ) we can define ***directional derivative***, in a given direction, of a differentiable function *f(x, y, z)* of three independent variables.



The direction is specified by the cosines of the three angles that it forms with the coordinate axes.

The directional derivative in the direction of unit vector $\vec{u}$, is given by

$f_x \cos\alpha + f_y \cos ß + f_z \cos y$ … … (**)

The unit vector $\vec{u}$ itself is ***specified*** as $\vec{u}$ **=< cos α,cos ß, cos y>**

Recalling the concept of scalar/ inner product of two vectors, from section 6.4 of Unit 6, we can write the expression (**) as scalar product of two vectors < *fx, fy, fz*> and <cos α,cos ß, cos y>.

In other words,

$< f_x, f_y, f_z > \cdot <\cos\alpha, \cos ß, \cos y> = f_x \cos\alpha + f_y \cos ß + f_z \cos y$ … … (**).

Hence, unit vector $\vec{u}$ in the direction of $\vec{v}$ **is** $\vec{u} = <-1/\sqrt{10}, 0, 3/\sqrt{10}>$

Thus,

$D\vec{u}\, f(x, y, z) = (-1/\sqrt{10})\,(f_x) + 0 + 3/\sqrt{10}\, f_z$

$= (-1/\sqrt{10})\,\{2z^2 - 2x\,yz\} + 3/\sqrt{10}\,\{4xz + 6y^2 z - x^2 y\} = \ldots$

Check Your Progress 2

For question Nos. 1, 2, and 3, find all the first order partial derivatives for the following functions.

1) $z = 8x / (x^2 + 5y)$

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2) $f(x; y; z) = x \sin(y)/ z^2$

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3) $z = \sqrt{(x^2 + \log(6x - 3y^2))}$

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4) Find $\partial^3 f/\partial y \partial x^2$, for $f(x, y) = e^{xy}$

....

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5) *For the function* $z = f(x, y) = 5x^3 y^4$, *calculate* $z_{xxx}, z_{xxy}, z_{xyx}, z_{xyy}, z_{yyy}$

....

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6) Find directional derivative for the function

$f(x, y) = x^2 \tan y + 3x \log y$ at the point $(1, \pi/3)$ along the unit vector $\vec{u}$ in the direction of $\theta = \pi/4$

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14.6 EXPLICIT FUNCTION FROM R^n TO R^m, JACOBIAN MATRIX, HESSIANS

We have already discussed the topic of functions from R^n to R^m in Section *13.5: Analysis of several variables* of Unit 13. Here, just recall the essential ideas about such functions.

14.6.1 Explicit Function from R^n to R^m

The essential idea, about a function from R^n to R^m, is explained through the example of, say,

$f: R^2 \rightarrow R^4$, such that

$f((x, y)) = (x+y^2, 3x^2-5, x. y, 3y^2)$,

which may be also considered as

$f((x, y)) = (f_1(x, y), f_2(x, y), f_3(x, y), f_4(x, y))$,

with

$f_1(x, y) = x+y^2$,

$f_2(x, y) = 3x^2 - 5$,

$f_3(x, y) = x.\ y$,

$f_4(x, y) = 3y^2$.

In general, for $\overline{x} = (x_1, \ldots, x_j, \ldots x_n) \in R^n$

a function $f: R^n \rightarrow R^m$ *may be considered as an m-tuple of the form*

$f(\overline{x}) = (f_1(\overline{x}), \ldots, f_i(\overline{x}), \ldots f_m(\overline{x}))$

(Please, note the last suffix on RHS is m in $f_m(\overline{x})$*, the suffix is not n)*

where each

f_j*:* $R^n \rightarrow R$*, is a function with codomain R, the set of real numbers.*

i. e, $f_j(\overline{x}) \in R$***, for each j =1, 2, …, m.***

14.6.2 Jacobian/Jacobian-Matrix

For the rest of the discussion, we assume that

the *nm* partial derivatives

$\partial f_i/\partial x_j$*, for i= 1,2, …, m* and *j =1, 2, …, n*

exist.

Definition: The Jacobian of the function $f: R^n \rightarrow R^m$ ***is the following*** $m \times n$ ***matrix of partial derivatives***

$J_f((\overline{x}) = (\partial f_i/\partial x_j)_{i=1,2,\ldots,m \text{ and } j=1,2,\ldots,n}$

In short, when function f and independent variable vector $\overline{x}$is understood to be known, then Jacobian is denoted just by J, and the (i, j)th entry is denoted by $J_{i,j} = \partial f_i/\partial x_j$.

The Jacobian **J may be alternatively represented as follows**

$$\mathbf{J} = \begin{bmatrix} \dfrac{\partial \mathbf{f}}{\partial x_1} & \cdots & \dfrac{\partial \mathbf{f}}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla f_1 \\ \vdots \\ \nabla f_m \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial f_m}{\partial x_1} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{bmatrix}$$

where, ∇f_i,the gradient of f_i, is the row-vector [$\partial f_i/\partial \boldsymbol{x_1}$**, …,** $\partial f_i/\partial \boldsymbol{x_j}$**, …,** $\partial f_i/\partial \boldsymbol{x_n}$**]**, for i= 1, 2, …, n.

Jacobian matrix J, is alternatively denoted as

Df, J_f, $\boldsymbol{\nabla f}$, $\frac{\boldsymbol{\partial}(\text{f1},\ldots,\text{fi},\ldots,\text{fm})}{\boldsymbol{\partial}(\boldsymbol{x1},\ldots,\boldsymbol{xj},\ldots,\boldsymbol{xn})}$, *or* $\boldsymbol{\partial}(\text{f1}, \ldots, \text{fi}, \ldots, \text{fm}) / \boldsymbol{\partial}(\boldsymbol{x1}, \ldots, \boldsymbol{xj}, \ldots, \boldsymbol{xn})$

Definition: Jacobian Determinant

If n= m, i. e. if the function f: $R^n \rightarrow R^n$, then the Jacobian matrix is a square matrix, and hence, its determinants can be defined. Such a determinant is called *Jacobian Determinant* or just Jacobian. Jacobian determinant is useful in evaluating multiple integrals.

(There is ambiguity here, as Jacobian matrix may also be called Jacobian)

Example 14.11

Let f: $R^2 \rightarrow R^2$ be defined as

$f(x, y) = (e^{2xy}, 2x^2 + 3y^2)$,

find the Jacobian J_f at the point (2, 1).

Let $f_1(x, y) = e^{2xy}$, **and** $f_2(x, y) = 2x^2 + 3y^2$

(Please note that '1' in f_1, here, does not indicate partial derivative)

$$\begin{bmatrix} \partial f_1/\partial x & \partial f_1/\partial y \\ \partial f_2/\partial x & \partial f_2/\partial y \end{bmatrix} = \begin{bmatrix} 2ye^{2xy} & 2xe^{2xy} \\ 4x & 6y \end{bmatrix}$$

Therefore, $J_f(2, 1) = \begin{bmatrix} 2e^4 & 4e^4 \\ 8 & 6 \end{bmatrix}$

14.6.3 Hessian

For a function $f: R^n \rightarrow R$,

which *maps* $\overline{x} = (x_1, x_2, \ldots, x_j, \ldots x_n) \in R^n$ *to* $f(x) \in R$

(Note, the codomain is just R, and not R^n)

the *Hessian is, an n* x n square matrix, *of all second order partial derivatives of f, i. e.*

$$\mathbf{H}_f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_1\,\partial x_n} \\[2.2ex] \dfrac{\partial^2 f}{\partial x_2\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} & \cdots & \dfrac{\partial^2 f}{\partial x_2\,\partial x_n} \\[2.2ex] \vdots & \vdots & \ddots & \vdots \\[2.2ex] \dfrac{\partial^2 f}{\partial x_n\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_n\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

In other words, the entry in the i^{th} row and j^{th} column of the matrix is

$$\partial^2 f / \partial x_i \partial x_j$$

Hessian matrix describes the local curvature of a function of many variables.

Example 14.12

Let f: $R^2 \to R$ be defined as

$$f(x, y) = e^{2xy}$$

find the Hessian H_f at the point (2, 1).

For $f(x, y) = e^{2xy}$,

$$\begin{bmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial f/\partial x & \partial f/\partial y \end{bmatrix} = \begin{bmatrix} 2ye^{2xy} & 2xe^{2xy} \\ 2ye^{2xy} & 2xe^{2xy} \end{bmatrix}$$

Therefore

$$H_f = \begin{bmatrix} \partial^2 f/\partial x^2 & \partial^2 f/\partial x\partial y \\ \partial^2 f/\partial y\partial x & \partial^2 f/\partial^2 y \end{bmatrix} = \begin{bmatrix} 4y^2e^{2xy} & (2+4xy)e^{2xy} \\ (2+4xy)e^{2xy} & 4x^2e^{2xy} \end{bmatrix}$$

where,

$\partial^2 f/\partial x^2$ etc denote second order partial derivative of fetc.**w. r. t. x.**

$$(H_f)_{(2,1)} = \begin{bmatrix} 4e^4 & 10e^4 \\ 10e^4 & \mathbf{16}\, e^4 \end{bmatrix}$$

Check Your Progress 3

1) Let f: $R^2 \to R^3$ be defined as

   $f(x, y) = (\sin x, 2xy, 3xy^2)$,

   find the Jacobian J_f at the point (π, $\pi/2$).

   ……………………………………………………………………………………………

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2) Let f: $R^3 \to R^3$ be defined as

   $f(x, y, z) = (2z + \sin x, 2xy - z^2, 3xy^2 - 3z)$,

   find the Jacobian J_f at the point (π, $\pi/2$, 1).

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3) Let f: $R^2 \to R$ be defined as

   $f((x, y)) = 3x^2 + 2y^3$

   find the Hessian H_f at the point (1, 2).

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14.7 MEAN VALUE THEOREM

The concept of *derivative* plays significant role in connecting the *global* to the *local*. For example, it is used in inferring a *global conclusion*, e. g., that a function is decreasing on an *interval* from *local information* that its derivative is negative at each *point* on an interval. On the other hand, the following theorem, called *Mean Value Theorem* relates the *global information* about the average rate of change of a function on an *interval to* the *local* information of the instantaneous rate of change at a *point* in the interval.

We just ***state, without proof,*** the Mean Value Theorem, first, for ***single*** variables in the next subsection, and then, for ***several*** variables in the subsequent subsection.

A little bit of history: A special case of this theorem for inverse interpolation of the sine was first described by *Parameshvara* (1380–1460), from the Kerala School of Astronomy and Mathematics in India, in his commentaries on *Govindasvāmi* and *Bhāskara* II. ***The theorem is now also known as Lagrange's Mean Value Theorem. (Source: Wikipedia)***

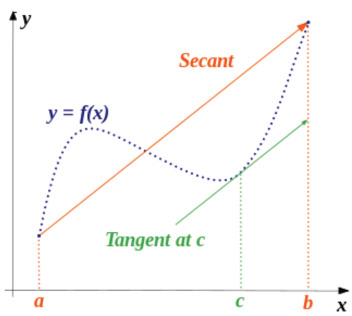
14.7.1 Mean Value Theorem for Single Variable

First, we give formally the statement of the theorem, and then give its geometric interpretation for its intuitive appeal.

Mean Value Theorem: For real numbers a and b, with $a < b$, ***if*** a function f: [a, b] → R, is continuous on the closed interval [a, b] and differentiable on the open interval *]a, b[****then*** there exists a point c in the open interval]a, b[such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

Geometric Interpretation of the Mean Value Theorem



If a curve of a function y= f(x) is *unbroken* (i. e. is continuous) on a closed interval [a, b], and has tangent (i. e. has derivative) at every point of the interval, except possibly at endpoints a and b, then there is at least one point (viz. c) at which the *tangent* to the arc is parallel to the *secant/line* joining its endpoints viz. *a* and *b*.

Example 14.13

Find the value of c of Lagrange's mean value theorem when

$f(y) = 2y^2 + 3y + 4$ in the interval [1, 2]

Solution:

As every *polynomial* is differentiable on its domain of definition, hence, continuous also, satisfying all the conditions of the Mean value theorem. Let c be the value in]1, 2[such that

$f'(c) = (f(b) - f(a)/(b-a)$, implying

$(f'(y) = 4y + 3)_{y=c} = 4c + 3 = \{f(2) - f(1)\}/(2 - 1) = (18 - 9)/1 = 9$

i.e. $4c + 3 = 9$ implying $c = 3/2 \in$]1, 2[.

14.7.2 Mean Value Theorem for Several Variables *(Optional reading)*

(Subsection 14.7.2 is optional reading; being included & discussed here only in view of the fact that the unit is about Calculus of ***several variables****)*

In order to generalize the Mean Value Theorem to the functions of n variables, we now consider the domain of function *f* as an *open* set G of R^n, instead of the open interval]a, b[of R, as was in the case of single variable. And the considered function is

$f: G \rightarrow R$, with $G \subseteq R^n$

(i. e., the inputs to the function are n-tuples $\overline{\boldsymbol{x}} = (x_1, \dots, x_i, \dots, xn)$*, with* $x_i \in R$*, for* $i = 1, 2, \dots, n$*, and* $\boldsymbol{f(\overline{x})} \in R$*).*

The statement of Mean Value Theorem, for several variables, then takes the form:

If a function $f: G \rightarrow R$ is differentiable at every point of the open set G, and if $\overline{\boldsymbol{x}}$, $\overline{\boldsymbol{y}} \in G$,

then, there exists a real number c between 0 and 1, such that

$f(\overline{\boldsymbol{y}}) - f(\overline{\boldsymbol{x}}) = [\nabla f((1-c)\,\overline{x} + c\,\overline{y})] \cdot (\overline{\boldsymbol{y}} - \overline{\boldsymbol{x}})$, **… … … (****)**

where ∇ is gradient; and **'.'** denotes 'dot product'.

The RHS of (****) may be further elaborated as follows:

i) The function f, has inputs which are n-tuples $\overline{\boldsymbol{x}} = (x_1, \dots, x_i, \dots, xn)$, *belonging to* $\boldsymbol{R^n}$*, but returns value* $\boldsymbol{f(\overline{x})}$ *which is an element of* $\boldsymbol{R}$.

ii) In specific case of (****) above, the input to f is the n-tuple $((1-c)\,\overline{x} + c\,\overline{y})$, with c between 0 and 1. In other words, the input to f corresponds to a point on the line joining $\overline{x}$ and $\overline{y}$ and the point is between $\overline{x}$ and $\overline{y}$.

iii) In n-dimensional space, $(\overline{y} - \overline{x})$ is the vector joining the end points of n-dimensional curve, under consideration

iv) from directional derivatives, under (***), we know ∇ f again is an n-tuple.

v) by (***a), ∇f. $(\bar{y}-\bar{x})$ is the directional derivative along n-tuple $(\bar{y}-\bar{x})$, where

vi) by combining (iii) with (v) above, and it is directional derivative of the vector corresponding to a point between$\bar{x}$ and $\bar{y}$.

vii) ∇f. $(\bar{y}-\bar{x})$, the directional derivative *(being derivative in the direction of $(\bar{y}-\bar{x})$)*, is the equation of the ***tangent*** parallel to $(\bar{y}-\bar{x})$, the vector joining the end points of the curve.

14.8 POLYNOMIAL APPROXIMATION: TAYLOR'S THEOREM, LINEAR & QUADRATIC APPROXIMATIONS

In this section, we discuss how to evaluate some functions—whose derivatives and/or partial derivatives exist up to some order—but which, in general, are difficult to evaluate ***in terms of rational numbers***. We can see ***how difficult the task of function evaluation, in general,*** is from the fact that evaluation of even of any of the so-called ***elementary functions*** like √x, $x^{2/3}$,e^x, log x, sin x, is not easy. For example, even for a small and simple value, say x=2, expressing any of the values √2, $2^{2/3}$,e^2, log 2, Sin 2, as rational numbers, is not easy.

The task is accomplished by *approximating* such a *function* in terms of some suitable *polynomials* depending upon the function under consideration. The question arises: ***Why in terms of polynomials?***

The answer lies in the fact *that evaluation of* ***polynomial*** *function is relatively easy, even when the degree of the polynomial may be large. This happens because, for example, $(2)^n$, n∈ N is rational, but e^2, or log (2) is not rational. In general, if x is rational then $(x)^n$, n∈ N, is rational, but each of e^x, or log (x) etc. is not rational.*

For a given function f: R→ R, (or in general f: R^n → R), there are various approaches for determining the *approximating* polynomial. However, we will be discussing only Taylor's approach, which is the most known &used, and some of its special cases,

Before we go into detailed discussion, we summarise (i) the issue, and (ii) the Taylor's approach to resolve the issue.

THE ISSUE:

If **the** values of the function, and its derivatives/ partial-derivatives, at some ***particular point x=a are known***, i. e. f(a) = $D^{(0)}$(a), Df(a) = f'(a)= $f^{(1)}$(a), D^2 f(a) = f''(a) = $f^{(2)}$(a), … D^k (f(a)) = $f^{(k)}$ (a), are known, where the derivatives Df, D^2f , … exist and are continuous in the domain of f.

Then to find out ***the value of f at a general point x*** in the domain, using the available knowledge of values at ***the particular point x=a***.

14.8.1 Taylor's Approach to Polynomial Approximation

i) Construct a polynomial, called **k^th^ Order Taylor's Polynomial**, **$P_{k,\ a}$**, involving the available knowledge about values of the derivatives ***$f^j(a)$*** at the point x=a, for j =0, 1, … k, as given by

The value ***f(x), at a general point x***, is given by the following ***Taylor's***

$$P_{k,\,a} = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k \qquad \textbf{(14.11)}$$

formula, involving Taylor's Polynomial:

$$f(x) = P_{k,\,a} + R_{k,\,a}(x), \qquad \ldots \text{(14.12)}$$

where, $R_{k,\,a}$ (x)**—which depends on k, the order of Taylor's polynomial, and a, the point at which the values of the function and its derivatives are known—**is called the remainder term.

The remainder term *$R_{k,\,a}$ (x)* denotes the ***error of approximation***, being the difference between the actual value at x *and* the value proposed by the

Remarks: ***The following about remainder term, including (a), (b), and (c) below—is mentioned only for the sake of completeness—and, may be ignored. We will not require the knowledge anywhere.***

The remainder term is expressed in different forms, including

a) Peano's form, in which

$$R_{k,\,a}(x) = h_{k,\,a}(x)\,(x-a)^k, \text{ such that}$$

$$\lim_{x\to a} \boldsymbol{h}_{k,\,a}(x) = 0.$$

b) Lagrange Mean-Value form of remainder

$$R_k(x) = \frac{f^{(k+1)}(\xi_L)}{(k+1)!}(x-a)^{k+1}$$

c) **Cauchy's form of remainder**

$$R_k(x) = \frac{f^{(k+1)}(\xi_C)}{k!}(x-\xi_C)^k(x-a)$$

approximating Taylor's Polynomial **$P_{k,\,a}$.**

For well-behaved functions f, we have$\lim_{x\to a}$ ***R***$_{k,\,a}$(x) = 0.

Now, the question arises: What is the justification/ validity ***by which we may be convinced to accept the Taylor's approach?***

The justification is provided by the well-known **Taylor's Theorem**, *which is stated below* **without proof.**

As can be seen from the text box above, many versions of the Taylor's theorem are expected, one version for each remainder form: Peano, Lagrange, Cauchy. Here, we state the theorem *only with general remainder form* ***$R_{k,\,a}$(x)****as given by* ***(#2)*** *above*. Many a time, when the point '*a'* is known from the context, we may drop *a*, and just write ***R_k (x)*** instead.

14.8.2 Taylor's Theorem

For integer k≥ 1, let the function f: R → R be k-times differentiable at the real number *a*. ***Then there exists*** a function R_k from Real numbers to Real numbers, such that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k +$$

R_k(x, a)

and

$\lim_{x\to a}$ **R_k**(x, a) =0,

where R_k (x, a) depends on x, a and the function f.

Another *Taylor Concept*, **probably, the final for the time being.**

14.8.3 Taylor's Series Or Taylor's Expansion

For some nice functions, like e^x and sin (x), a function may be ***infinitely*** differentiable at some point(s) of Domain.

For such functions f: R → R, we have the following expansion for the general value of the function f

f(x) =

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots, \qquad \textbf{(\#14.13)}$$

In compact summation form, it may be denoted as

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n,$$

where, n! denotes factorial of n, $f^{(n)}(a)$ denotes the value of n^{th} derivative at the point x=a, $f^{(0)}$= f(a), $(x – a)^0$=1, and

0! =1